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Notation. $\mathbb{Z}$ (resp. $\mathbb{Q}, \mathbb{R}, \mathbb{C}$) denotes the set of integers (resp. rational numbers, real numbers, complex numbers).

1 Review of number theory

1.1 The Euclidean Algorithm

Definition 1.1. Let a and b be given integers, with at least one of them different from zero. The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

Example 1.2. For $a = 35^27$ and $b = 25^2711$, we know that $\gcd(a, b) = 5^27$.

How to find $\gcd(a, b)$? Note that prime factorization is hard. Therefore we will use the Euclidean algorithm to find $\gcd(a, b)$ without prime factorization. Let introduce the division algorithm and the Euclidean algorithm.

[The Division Algorithm]

- **Input** : $a, b \in \mathbb{Z}$ and $b \neq 0$.
- **Output** : $a = qb + r$ and $0 \leq r < |b|$.

[The Euclidean Algorithm]

- **Input** : nonzero integers a, b ($|a| \geq |b|$).
 - **Output** : $\gcd(a, b)$ of the input integers a and b .
1. Let $r_0 = b$.
 2. By the division algorithm, there exists r_1, q_1 such that $a = q_1b + r_1$ and $0 \leq r_1 < |b|$.
 3. If $r_i \neq 0$, then there exists q_{i+1}, r_{i+1} such that $r_{i-1} = q_{i+1}r_i + r_{i+1}$ and $0 \leq r_{i+1} < r_i$. (Using the division algorithm)
 4. If $r_i = 0$, then $\gcd(a, b) = r_{i-1}$.

Example 1.3. 1. Find the greatest common divisor of 15 and 3.

$$15 = 3 \times 5 + 0$$

Therefore $\gcd(15, 3) = r_0 = 3$.

2. Find the greatest common divisor of 15 and 6.

$$15 = 6 \times 2 + 3$$

$$6 = 3 \times 2 + 0$$

Therefore $\gcd(15, 6) = r_1 = 3$.

For some nonzero integers a, b ($|a| \geq |b|$), apply the Euclidean algorithm to integers a, b . Assume that $r_3 = 0$. In other words,

$$a = bq_1 + r_1, b = r_1q_2 + r_2, r_1 = r_2q_3 + 0.$$

We know that $r_1 = a - bq_1$ and a, b are multiple of $\gcd(a, b)$. So r_1 is multiple of $\gcd(a, b)$. Similarly, we know that $r_2 = b - r_1q_2$ and b, r_1 are multiple of $\gcd(a, b)$. So r_2 is multiple of $\gcd(a, b)$. The remaining proof of $r_2 = \gcd(a, b)$ is omitted.

1.2 Solving the Diophantine Equation $ax + by = c$

Let $a, b, c \in \mathbb{Z}$. How to solve $ax + by = c$? More specifically, we are curious about ‘ $ax + by = c$ has a solution?’ and ‘Find the all solutions of $ax + by = c$ ’. At first, let’s talk about ‘ $ax + by = c$ has a solution?’.

Theorem 1.4. *For nonzero integers a and b , let $T = \{ax + by | x, y \in \mathbb{Z}\}$. Then T is the set of all multiple of $\gcd(a, b)$.*

For example, $\{6x + 9y\} = \{3n | n \in \mathbb{Z}\}$. Remark $ax + by = c$ has an integral solution if and only if $c | \gcd(a, b)$. Therefore there exists $x_0, y_0 \in \mathbb{Z}$ such that $ax_0 + by_0 = \gcd(a, b)$. If $c = \gcd(a, b)m$, then $a(mx_0) + b(my_0) = c$. If $\gcd(a, b) \nmid c$, then $ax + by = c$ has no integral solution. If we can find the root of $ax + by = c$ when c is $\gcd(a, b)$, we can also find the root when c is a multiple of $\gcd(a, b)$. Let introduce an algorithm to find an integral solution of $ax + by = c$ such that $c = \gcd(a, b)$.

- **Input** : $ax + by = c$ such that $c = \gcd(a, b)$ and $|a| \geq |b|$.
 - **Output** : an integral solution x_0, y_0 such that $ax_0 + by_0 = c$.
1. Let $r_0 = b$ and $m_1 = 1$.
 2. By the division algorithm, there exists r_1, q_1 such that $r_1 = a - q_1b$ and $0 \leq r_1 < |b|$. Let $n_1 = -q_1$. Then $r_1 = m_1a + n_1b$.

3. By the division algorithm, there exists r_2, q_2 such that $r_0 = q_2 r_1 + r_2$ and $0 \leq r_2 < r_1$. Let $m_2 = -q_2$ and $n_2 = 1 + q_1 q_2$. Then $r_2 = m_2 a + n_2 b$.
4. If $r_i \neq \gcd(a, b)$, then there exists r_{i+1}, q_{i+1} such that $r_{i-1} = q_{i+1} r_i + r_{i+1}$ and $0 \leq r_{i+1} < r_i$. Let $m_{i+1} = m_{i-1} - q_{i+1} m_i$ and $n_{i+1} = n_{i-1} - q_{i+1} n_i$. Then $r_{i+1} = m_{i+1} a + n_{i+1} b$.
5. If $r_i = \gcd(a, b)$, then $x_0 = m_i$ and $y_0 = n_i$.

Exercise 1.5. Find $\gcd(65, 105)$ by using prime factorization.

Exercise 1.6. Find $\gcd(a, b)$ by Euclidean algorithm.

1. $\gcd(65, 105)$

2. $\gcd(202, 98)$

Exercise 1.7. Complete the following statement.

“ $ax + by = c (a, b, c \in \mathbb{Z})$ has an integral solution if and only if ”

Exercise 1.8. Find an integral solution such that $56x + 72y = 8$.

Proof. Replace x and y . Find an integral solution such that $72x + 56y = 8$.

1. $r_0 = 56, m_1 = 1$.

2. $r_1 = 16 = 72 - 1 \times 56, q_1 = 1$.

3. $r_2 = 8 = 56 - 3 \times 16, q_2 = 3, m_2 = -3, n_2 = 4$. Then $r_2 = -3 \times 72 + 4 \times 56$.

4. $0 = r_3 = 16 - 2 \times 8$.

Therefore we have $8 = -3 \times 72 + 4 \times 56$.

Replace x and y again, then $(x, y) = (4, -3)$ is an integral solution. □

1.3 Find all of integral solutions of $ax + by = c (a, b, c \in \mathbb{Z})$

Notation. When we write solution, it means integer solution.

Note that $ax + by = c$ has a solution if and only if $\gcd(a, b) | c$.

Observe the following formula. If $ax_0 + by_0 = c$ and $ax_n + by_n = 0$, then $a(x_0 + x_n) + b(y_0 + y_n) = c$. Therefore $(x, y) = (x_0 + x_n, y_0 + y_n)$ is another solution of $ax + by = c$. If $ax_0 + by_0 = c$ and $ax'_n + by'_n = 0$, then $a(x_0 - x'_0) + b(y_0 - y'_0) = 0$. Therefore $(x, y) = (x_0 - x'_0, y_0 - y'_0)$ is solution

of $ax + by = 0$. Note that find a solution of homogeneous equation ($ax + by = 0$) is important for finding solution of $ax + by = c$.

Theorem 1.9. *Let $a, b, c \in \mathbb{Z}$. Assume that $(x, y) = (x_0, y_0)$ is an integral solution of $ax + by = c$. Then (x', y') is a solution of $ax + by = c$ if and only if*

$$x' = x_0 + x_n, y' = y_0 + y_n$$

for some $(x_n, y_n) \in \mathbb{Z} \times \mathbb{Z}$ such that $ax_n + by_n = 0$.

To find all solutions of $ax + by = c$, we need to find all solutions of $ax + by = 0$. Let's find all integral solutions of $ax + by = 0$. Let $a = a' \gcd(a, b)$ and $b = b' \gcd(a, b)$. Then $(a', b') = 1$.

$$\begin{aligned} ax + by &= 0 \\ \Rightarrow ax &= -by \\ \Rightarrow \gcd(a, b)a'x &= -\gcd(a, b)b'y \\ \Rightarrow a'x &= -b'y \\ \Rightarrow x &= b'n, y = -a'n \text{ for some } n \in \mathbb{Z} (\cdot (a', b') = 1) \end{aligned}$$

Theorem 1.10. *Every solution of equation $ax + by = 0$ is given by*

$$x = \frac{b}{\gcd(a, b)}n, y = -\frac{a}{\gcd(a, b)}n$$

where n is an arbitrary integer.

To summarize how to find all solutions of $ax + by = c$.

1. Find a particular solution (x_0, y_0) such that $ax_0 + by_0 = c$.
2. General solution of $ax + by = c$ is represented by $x = x_0 + \frac{b}{\gcd(a, b)}n, y = y_0 - \frac{a}{\gcd(a, b)}n$ for some $n \in \mathbb{Z}$.

Exercise 1.11. *Find all integral solutions of $ax + by = 0$.*

1. $120x + 24y = 0$.
2. $256x + 30y = 0$.

Exercise 1.12. *Find a particular solution of $ax + by = c$.*

- $172x + 20y = 1000$.
- $278x + 102y = 2$.

Exercise 1.13. Find all integral solutions.

1. $172x + 20y = 1000$.
2. $278x + 102y = 2$.

Proof. 1) By Exercise 1.12., we know that $4 = 172 \times 500 - 20 \times 4250$ and $\gcd(172, 20) = 4$. So every integral solution of $172x + 20y = 1000$ is represented by $(x, y) = (500 + 5n, -4250 - 43n)$ for some $n \in \mathbb{Z}$.

□

1.4 Modulo arithmetic

Definition 1.14. Let n be a fixed positive integer. Two integers a and b are said to be congruent modulo n ,

$$a \equiv b \pmod{n}$$

if $n \mid a - b$.

Remark 1. Suppose that $a = q_1n + r_1$ ($0 \leq r_1 < |n|$) and $b = q_2n + r_2$ ($0 \leq r_2 < |n|$). Then

$$a \equiv b \pmod{n} \Leftrightarrow r_1 = r_2.$$

Example 1.15. 1. $12 \equiv 7 \pmod{5}$

2. $23 \equiv 7 \pmod{8}$

3. $1000 \equiv 4 \pmod{7}$

Theorem 1.16. Let $n > 1$ be fixed and $a, b, c, d \in \mathbb{Z}$. Then the following properties hold:

1. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$ and $ac \equiv bd \pmod{n}$.
2. Suppose that $\gcd(t, n) = 1$. If $ta \equiv tb \pmod{n}$, then $a \equiv b \pmod{n}$.

Example 1.17. 1. Suppose that $3 \cdot 6 \equiv 3 \cdot x \pmod{35}$. Then $\gcd(3, 35) = 1$. So $x \equiv 6 \pmod{35}$.

2. $3 \cdot 4 \equiv 3 \cdot 8 \pmod{6}$. But $4 \not\equiv 8 \pmod{6}$ ($\because (3, 6) \neq 1$).

Theorem 1.16 2).

$$ta \equiv tb \pmod{n} \Leftrightarrow n \mid (ta - tb) = t(a - b) \Leftrightarrow n \mid (a - b) (\because (n, t) = 1)$$

□

Example 1.18. 1.

$$\begin{aligned} 10000 \cdot 73 &\equiv 4 \cdot 3 \pmod{7} \\ &\equiv 12 \pmod{7} \\ &\equiv 5 \pmod{7} \end{aligned}$$

$$2. \quad 273 \cdot 107 \equiv 3 \cdot 2 \equiv 1 \pmod{5}$$

$$3. \quad 5^3 \equiv 25 \cdot 5 \equiv 1 \cdot 5 \equiv 5 \pmod{12}$$

Consider the formula $a^x \equiv y \pmod{n}$. Suppose that we know that a, x and find y . It's a relatively easy problem. But suppose that we know that a, y and find x . It's a hard problem. The fact that this problem is hard makes it difficult to solve the RSA password.

Example 1.19. 1. $3^4 \equiv (3^2)^2 \equiv 9^2 \equiv 81 \equiv 4 \pmod{11}$

$$2. \quad 3^8 \equiv (3^4)^2 \equiv 4^2 \equiv 5 \pmod{11}$$

$$3. \quad 3^{16} \equiv (3^8)^2 \equiv 5^2 \equiv 3 \pmod{11}$$

$$4. \quad 3^{32} \equiv (3^{16})^2 \equiv 3^2 \equiv 9 \pmod{11}$$

$$5. \quad \text{Calculate } 3^{40} \pmod{11}. \text{ Note that } 40 = 2^5 + 2^3.$$

$$\begin{aligned} 3^{40} &\equiv 3^{32} 3^8 \pmod{11} \\ &\equiv 9 \times 5 \pmod{11} \\ &\equiv 1 \pmod{11} \end{aligned}$$

Notice that we only did 6 multiplications.

Let $r \in \mathbb{Z}_{>0}$ and $r = b_n 2^n + \cdots + b_1 2 + b_0$ ($b_i \in \{0, 1\}$). Calculate $a^r \pmod{m}$ using 2-digit expression.

$$\begin{aligned} a^r &\equiv a^{b_n 2^n + \cdots + b_1 2 + b_0} \pmod{m} \\ &\equiv a^{b_n 2^n} \cdots a^{b_1 2} a^{b_0} \pmod{m} \end{aligned}$$

Let's count the number of multiplication of calculate above formula.

1. Calculate $a^{2^n}, \dots, a^2$. It requires n times multiplication.
2. Calculate $a^{b_n 2^n} \times \dots \times a^{b_0}$. It requires n or fewer times multiplication.

Theorem 1.20. *It is possible to calculate $a^N \pmod{n}$ using only $2\lfloor \log_2 r \rfloor$ multiplication.*

For example, suppose a computer does 1 billion modulo n multiplication per a second. Compute $a^N \pmod{n}$ when $N = 10^{20}$. By multiplication itself 10^{20} times take about 3000 years. But using the above method, multiplication $2\lfloor \log_2 10^{20} \rfloor$ times take 0.1 micro seconds.

(Algorithm for powers $\pmod{n}$)

- **Input** : $x \in \mathbb{Z}, n, N \in \mathbb{Z}_{>0}$ and $n \leq 2$.
- **Output** : $\text{power}(x, N)$ (integer such that $x^N \equiv \text{power}(x, N) \pmod{n}$ and $0 \leq \text{power}(x, N) < n$)

(a recursive algorithm)

$$\text{power}(x, N) = \begin{cases} \text{return } x, & \text{if } N = 1 \\ \text{return } 0 \leq r < n \text{ s.t. } r \equiv \text{power}\left(x, \frac{N}{2}\right)^2 \pmod{n}, & \text{if } N \text{ even} \\ \text{return } 0 \leq r < n \text{ s.t. } r \equiv x \text{power}\left(x, \frac{N-1}{2}\right)^2 \pmod{n}, & \text{if } N \text{ odd and } N \geq 2 \end{cases} \quad (1)$$

Exercise 1.21. Find an integral solution of $24x + 9y = 3$.

Exercise 1.22. Compute $1001 \cdot 409 \pmod{9}$.

Exercise 1.23. Complete the following statement.

It is possible to calculate $a^N \pmod{n}$ using only times multiplications."

Exercise 1.24. Compute $2^{341} \pmod{340}$ by using the 2-digit expression.

Proof. Note that $341 = 101010101_{(2)}$.

$$\begin{aligned} 2^{341} &\equiv 2^{101010101_{(2)}} \pmod{340} \\ &\equiv 2^{256} 2^{64} 2^{16} 2^4 \pmod{340} \end{aligned}$$

We know that $2 \equiv 2(mod340)$, $2^2 \equiv 4(mod340)$, $2^4 \equiv 16(mod340)$, $2^8 \equiv 256(mod340)$, $2^{16} \equiv 65536 \equiv 256(mod340)$. Similarly $2^{256} \equiv 2^{128} \equiv 2^{64} \equiv 2^{32} \equiv 256(mod340)$.

$$\begin{aligned}
 2^{341} &\equiv 2^{101010101_{(2)}}(mod340) \\
 &\equiv 2^{256}2^{64}2^{16}2^4(mod340) \\
 &\equiv 256 \times 256 \times 256 \times 16 \times 2(mod340) \\
 &\equiv 256 \times 16 \times 2(mod340) \\
 &\equiv 8192(mod340) \\
 &\equiv 32(mod340)
 \end{aligned}$$

Therefore $2^{341} \equiv 32(mod340)$. □

Exercise 1.25. Compute $2^{341}(mod340)$ by the algorithm step by step.

Proof.

$$\begin{aligned}
 power(2, 341) &= 2(power(2, 170))^2 \\
 &= 2((power(2, 85))^2)^2 \\
 &= 2((2(power(2, 42))^2)^2)^2 \\
 &= 2((2(power(2, 42))^2)^2)^2 \\
 &= 2((2((power(2, 21))^2)^2)^2)^2 \\
 &= 2((2((2(power(2, 10))^2)^2)^2)^2)^2 \\
 &= 2((2((2((power(2, 5))^2)^2)^2)^2)^2)^2 \\
 &= 2((2((2((2(power(2, 2))^2)^2)^2)^2)^2)^2)^2 \\
 &= 2((2((2((2(4)^2)^2)^2)^2)^2)^2)^2 \\
 &= 2((2((2((32)^2)^2)^2)^2)^2)^2 \\
 &= 2((2((2(1024)^2)^2)^2)^2)^2 \\
 &= 2((2((2(4)^2)^2)^2)^2)^2 \\
 &= 2((2((32)^2)^2)^2)^2
 \end{aligned}$$

$$\begin{aligned}
 &= 2((2(1024)^2)^2)^2 \\
 &= 2((2(4)^2)^2)^2 \\
 &= 2((32)^2)^2 \\
 &= 2(1024)^2 \\
 &= 2(4)^2 \\
 &= 32
 \end{aligned}$$

□

1.5 The inverse of an integer modulo n

Note that $3 \times 7 \equiv 1(\text{mod}5)$. Then 3 looks like a multiplicative inverse of $7 \text{mod} 5$.

Definition 1.26. For $a \in \mathbb{Z}$, if $b \in \mathbb{Z}$ satisfies $ab \equiv 1(\text{mod}n)$, then b is called the inverse of a modulo n , denoted by $a^{-1}(\text{mod}n)$.

Example 1.27. 1. Note that $3 \cdot 4 \equiv 1(\text{mod}11)$. Then 3 is inverse of $4(\text{mod}11)$.

2. Note that $5 \cdot 5 \equiv 1(\text{mod}12)$. Then 5 is inverse of $5(\text{mod}12)$.

3. Suppose that there exists $x \in \mathbb{Z}$ such that a inverse of 2 mod 10. Then

$$2x \equiv 1(\text{mod}10) \Rightarrow 10|(2x - 1).$$

However $2x - 1$ is odd and so it is impossible. Therefore 2 has no inverse mod 10.

Let's fix n . Which integer a is invertible?

Theorem 1.28. For $a, b, n \in \mathbb{Z}$, if $a \equiv b(\text{mod}n)$, then $\gcd(a, n) = \gcd(b, n)$.

Proof. Assume that $a \equiv b(\text{mod}n)$. Then $n|(a - b)$. So $a - b = nq$ for some $q \in \mathbb{Z}$. Therefore $\gcd(b, n)|nq + b = a$ and $\gcd(a, n)|a - nq = b$. We know that $\gcd(b, n)|n$ and $\gcd(a, n)|n$. In fact, for $a, b, m \in \mathbb{Z}$, if $m|a$ and $m|b$, then $m|\gcd(a, b)$. Therefore $\gcd(b, n)|\gcd(a, n)$ and $\gcd(a, n)|\gcd(b, n)$. Note that if $A, B \in \mathbb{Z}_{>0}$, $A|B$, and $B|A$, then $A = B$. Therefore $\gcd(a, n) = \gcd(b, n)$. □

Theorem 1.29. For $a, n \in \mathbb{Z}$, integer a is an invertible mod n if and only if $\gcd(n, a) = 1$.

Proof.

$$\begin{aligned}
 &\text{Integer } a \text{ is an invertible mod } n \\
 &\Leftrightarrow \exists x \in \mathbb{Z} \text{ such that } ax \equiv 1 \pmod{n} \\
 &\Leftrightarrow n \mid ax - 1 \\
 &\Leftrightarrow ny = ax - 1 \text{ for some } y \in \mathbb{Z} \\
 &\Leftrightarrow ax + ny = 1 \text{ for some } y \in \mathbb{Z} \\
 &\gcd(a, n) = 1
 \end{aligned}$$

□

Theorem 1.30. For $a, n \in \mathbb{Z}$, let a be an invertible mod n . If $b \equiv a \pmod{n}$, then b is an invertible mod n .

Proof. Since $b \equiv a \pmod{n}$, we know that $\gcd(n, a) = \gcd(n, b)$. Since a invertible mod n , we have $\gcd(a, n) = 1$. Therefore $\gcd(b, n) = 1$. Finally, b is invertible mod n . □

For $n \in \mathbb{Z}_{>0}$, the invertible of a modulo n is associated with $a \pmod{n}$. Note that if p is a prime, then every $a \in \mathbb{Z}$ such that $p \nmid a$ is invertible.

Example 1.31. We know that 1, 5, 7, and 11 is invertible mod 12. Therefore $12k+1, 12k+5, 12k+7$, and $12k+11$ are invertible mod 12.

Theorem 1.32. Let $n \in \mathbb{Z}_{>0}$, $n \geq 2$, and $a \in \mathbb{Z}$.

1. If a has an inverse modulo n , then its inverse is unique modulo n . In other words, if $ab \equiv 1 \pmod{n}$ and $ac \equiv 1 \pmod{n}$, then $b \equiv c \pmod{n}$.
2. $(a^{-1})^{-1} = a$ if a is an invertible mod n .

Proof. 1. Assume that $ab \equiv ac \equiv 1 \pmod{n}$. Then $ab - ac \equiv a(b - c) \equiv 0 \pmod{n}$. And so $n \mid a(b - c)$. Since $\gcd(n, a) = 1$, $n \mid b - c$. Therefore $b \equiv c \pmod{n}$.

2. Since $a \cdot a^{-1} \equiv 1 \pmod{n}$, a is inverse of a^{-1} . Therefore $(a^{-1})^{-1} = a$.

□

[Inverse mod n Algorithm]

- **Input** : nonzero integers a, n such that $\gcd(a, n) = 1$
- **Output** : $a^{-1}(\text{mod } n)$

Find $x, y \in \mathbb{Z}$ such that $ax + ny = 1$ by Euclidean algorithm. Then $a^{-1} \equiv x(\text{mod } n)$.

Exercise 1.33. Which numbers $0 \leq a \leq 19$ are invertible modulo 20?

Exercise 1.34. Prove that if a is invertible modulo n , then its inverse is unique.

Exercise 1.35. a is invertible modulo n if and only if $\gcd(a, n) = 1$. Fill the (A) and (B).

Exercise 1.36. By the Euclidean algorithm, find $14^{-1}(\text{mod } 23)$.

1.6 Find a solution of $ax \equiv b(\text{mod } n)$

Theorem 1.37. Let $n \in \mathbb{N}$, $n \geq 2$, and $a, b \in \mathbb{Z}$. If $\gcd(a, n) = 1$, then $ax \equiv b(\text{mod } n)$ has a unique solution.

Proof. Since $\gcd(a, n) = 1$, there exists $a^{-1}(\text{mod } n)$.

$$\begin{aligned} ax &\equiv b(\text{mod } n) \\ \Leftrightarrow a^{-1}(ax) &\equiv a^{-1}b(\text{mod } n) \\ \Leftrightarrow x &\equiv a^{-1}b(\text{mod } n) \end{aligned}$$

Therefore $ax \equiv b(\text{mod } n)$ has a unique solution $x \equiv a^{-1}b(\text{mod } n)$.

□

Example 1.38. Find the solution of $3x \equiv 2(\text{mod } 8)$. Since $\gcd(3, 8) = 1$, by theorem 1.37, $3x \equiv 2(\text{mod } 8)$ has a unique solution. And then $x \equiv 3^{-1}2 \equiv 3 \times 2 \equiv 6(\text{mod } 8)$.

Remark 2. 1. The formula $3x \equiv 0(\text{mod } 6)$ has solutions $x \equiv 0, 2, 4(\text{mod } 6)$ not a unique solution.

2. Let a^{-1} be the inverse of a modulo n . Define $a^{-n} := a^{-1} \cdots a^{-1}$, n times multiple of a^{-1} . For $m \in \mathbb{N}$, note that $a^{-1}a^m \equiv a^{m-1}(\text{mod } n)$. Therefore $a^m a^{m'} \equiv a^{m+m'}(\text{mod } n)$ for all $m, m' \in \mathbb{Z}$.

1.7 Chinese remainder theorem

Let $x \equiv 4(\text{mod } 7)$, $x \equiv 9(\text{mod } 11)$, and $x \equiv 3(\text{mod } 13)$. Find the integer such that $T_1 \equiv 4(\text{mod } 7)$, $T_2 \equiv 0(\text{mod } 7)$, $T_3 \equiv 0(\text{mod } 7)$, $T_1 \equiv 0(\text{mod } 11)$, $T_2 \equiv 9(\text{mod } 11)$, $T_3 \equiv 0(\text{mod } 11)$, $T_1 \equiv 0(\text{mod } 13)$, $T_2 \equiv$

$0(\text{mod}13)$ and $T_3 \equiv 3(\text{mod}13)$. Then $T_1 + T_2 + T_3$ is a solution of the system. Let $T_1 = a_1 \cdot 11 \cdot 13$, $T_2 = a_2 \cdot 7 \cdot 13$ and $T_3 = a_3 \cdot 7 \cdot 11$ for some integer a_1, a_2, a_3 . Then $T_2 \equiv 0(\text{mod}7), T_3 \equiv 0(\text{mod}7)$, $T_1 \equiv 0(\text{mod}11), T_3 \equiv 0(\text{mod}11)$, $T_1 \equiv 0(\text{mod}13)$ and $T_2 \equiv 0(\text{mod}13)$. Find $a_1, a_2, a_3 \in \mathbb{Z}$ such that

$$a_1 \cdot 11 \cdot 13 \equiv 4(\text{mod}7), a_2 \cdot 7 \cdot 13 \equiv 9(\text{mod}11), a_3 \cdot 7 \cdot 11 \equiv 3(\text{mod}13).$$

Then $a_1 \equiv (11 \cdot 13)^{-1} \cdot 4(\text{mod}7)$, $a_2 \equiv (7 \cdot 13)^{-1} \cdot 9(\text{mod}11)$, $a_3 \equiv (7 \cdot 11)^{-1} \cdot 3(\text{mod}13)$. So $a_1 \equiv 6(\text{mod}7), a_2 \equiv 3(\text{mod}11), a_3 \equiv 10(\text{mod}13)$. Finally

$$x \equiv T_1 + T_2 + T_3 \equiv 6 \cdot 11 \cdot 13 + 3 \cdot 7 \cdot 13 + 10 \cdot 7 \cdot 11 \equiv 900(\text{mod}7 \cdot 11 \cdot 13).$$

Theorem 1.39 (Chinese remainder theorem). *Let $m_1, \dots, m_n$ be the pairwise relatively prime positive integers. For $a_i \in \mathbb{Z}, 1 \leq i \leq n$, the system $x \equiv a_1(\text{mod}m_1), \dots, x \equiv a_n(\text{mod}m_n)$ has a unique solution modulo $m_1 \cdots m_n$.*

Proof. (Uniqueness)

Assume that $x, y \in \mathbb{Z}$ are solutions of the above system. Then $x \equiv y \equiv a_i(\text{mod}m_i)$ for all i . So $x - y \equiv 0(\text{mod}m_i)$ for all i . In other words, $m_i | (x - y)$ for all i . Therefore $\text{lcm}(m_1, \dots, m_n) | (x - y)$. Note that if $m_1, \dots, m_n$ are the pairwise relatively prime positive integers, then $\text{lcm}(m_1, \dots, m_n) = m_1 \cdots m_n$. So $m_1 \cdots m_n | (x - y)$. Therefore $x \equiv y(\text{mod}m_1 \cdots m_n)$. $\square$

[Chinese remainder theorem]

- **Input** : pairwise relatively prime positive integers $m_1, \dots, m_n$ and $a_1, \dots, a_n \in \mathbb{Z}$
 - **Output** : a solution of the system $x \equiv a_1(\text{mod}m_1), \dots, x \equiv a_n(\text{mod}m_n)$
1. Define $M = m_1 \cdots m_n$ and $M_i = M/m_i$.
 2. Find $N_i \in \mathbb{Z}$ such that N_i is inverse of M_i modulo m_i .
 3. A solution of the above system is $x \equiv a_1 N_1 m_1 + \cdots + a_n N_n m_n (\text{mod}M)$.

Exercise 1.40. Solve $13x \equiv 4(\text{mod}80)$.

Exercise 1.41. Solve $3x \equiv 0(\text{mod}18)$.

Exercise 1.42. Solve the system by the algorithm CRT.

1. $x \equiv 2(\text{mod}3), x \equiv 4(\text{mod}5)$ and $x \equiv 6(\text{mod}7)$
2. $x \equiv 2(\text{mod}3), x \equiv 4(\text{mod}5)$ and $x \equiv 2(\text{mod}7)$

Exercise 1.43. Solve the equation $x^2 + 1 \equiv 0(\text{mod}85)$.

1.8 Euler ϕ function

Definition 1.44. Given a positive integer n , let $\phi(n)$ denote the number of a such that $1 \leq a \leq n$ and $\gcd(a, n) = 1$. The function ϕ is called the Euler *phi* function.

Remark 3. For a positive integer n , the integer $\phi(n)$ denotes the number of invertible a ($1 \leq a \leq n$) modulo n .

Example 1.45. Let's calculate $\phi(12)$. With $\gcd(a, 12) = 1$, an a must not be a multiple of 2 and 3. So possible a such that $1 \leq a \leq 12$ and $\gcd(a, 12) = 1$ is 1, 5, 7, 11. Therefore $\phi(12) = 4$.

How to find the value of $\phi(n)$ for $n \in \mathbb{N}$? We consider this question as the following steps :

1. Calculate $\phi(p^n)$ for prime p and $n \in \mathbb{Z}_{>0}$
2. Calculate $\phi(p_1^{k_1} \cdots p_m^{k_m})$.

For example, $\phi(9) = \phi(3^2)$. Note that $\gcd(a, 9) = 1$ if and only if $\gcd(a, 3) = 1$. In other words, $\gcd(a, 9) > 1$ if and only if $a = 3l$ for $l \in \mathbb{N}$. So 3, 6, and 9 are the only number for which $\gcd(a, 9) > 1$ and $1 \leq a \leq 9$. So $\phi(9) = 9 - 3 = 6$.

Theorem 1.46. Let p be a prime and $k \in \mathbb{N}$. Then $\phi(p^k) = p^{k-1}(p - 1)$.

Proof. Note that $\gcd(n, p^k) = 1$ if and only if $\gcd(n, p) = 1$. Therefore $\phi(p^k)$ is the number of a ($1 \leq a \leq p^k$) such that $\gcd(a, p) = 1$. So $p^k - \phi(p^k)$ is the number of a ($1 \leq a \leq p^k$) such that $\gcd(a, p) = p$. In other words, it is the number of the multiple of p below p^k . It is the number of $p, 2p, \dots, p^{k-1}p = p^k$. So $p^k - \phi(p^k) = p^{k-1}$, $\phi(p^k) = p^k - p^{k-1} = p^{k-1}(p - 1)$. $\square$

Let $n = p_1^{a_1} \cdots p_m^{a_m}$ be the prime factorization of the positive integer n .

Theorem 1.47. Let m and n be relatively prime positive integers. Then $\phi(mn) = \phi(m)\phi(n)$.

Since above theorem, we know that $\phi(p_1^{a_1} \cdots p_m^{a_m}) = \phi(p_1^{a_1}) \cdots \phi(p_m^{a_m})$. For example, calculate $\phi(540)$. Note that $540 = 2^2 3^3 5$. Then $\phi(540) = \phi(2^2)\phi(3^3)\phi(5) = 2(2-1) \cdot 3^2(3-1) \cdot (5-1) = 144$.

Remark 4. Note that $\phi(3)\phi(3) = 2 \times 2 = 4$ and $\phi(3^2) = 3(3-1) = 6$. The above theorem is false without assumption $(m, n) = 1$.

Theorem 1.48. Let $n = p_1^{a_1} \cdots p_m^{a_m}$ be the prime factorization of the positive integer n . Then

$$\phi(n) = \prod_{i=1}^k p_i^{a_i-1}(p_i - 1) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right)$$

Proof.

$$\begin{aligned}
 \phi(n) &= \phi(p_1^{a_1} \cdots p_k^{a_k}) \\
 &= \phi(p_1^{a_1}) \cdots \phi(p_k^{a_k}) \\
 &= p_1^{a_1-1}(p_1 - 1) \cdots p_k^{a_k-1}(p_k - 1) \\
 &= np_1^{-1}(p_1 - 1) \cdots p_k^{-1}(p_k - 1) \\
 &= n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right)
 \end{aligned}$$

□

For given $n, a \in \mathbb{N}$, find $m \in \mathbb{N}$ such that $a^m \equiv 1 \pmod{n}$. We will study about above question.

Definition 1.49. Assume that $a, n \in \mathbb{N}$ such that $\gcd(a, n) = 1$. The order of a modulo n is the least positive integer k such that $a^k \equiv 1 \pmod{n}$.

Example 1.50. 1. The order of 1 modulo n is 1.

2. Think about order of 2 mod 5.

n	1	2	3	4
$2^n \pmod{5}$	2	4	3	1

Therefore the order of 2 mod 5 is 4.

3. Think about order of 2 mod 11.

n	1	2	3	4	5	6	7	8	9	10
$2^n \pmod{11}$	2	4	8	5	10	9	7	3	6	1

Therefore the order of 2 mod 11 is 10.

Theorem 1.51. Let a and n be positive integers such that $\gcd(a, n) = 1$.

Proof. Let t be the order of a modulo n .

$\Rightarrow$ Assume that $a^m \equiv 1 \pmod{n}$. Then $m = qt + r$ for some integer q and $0 \leq r < t$. We know that $1 \equiv a^m \equiv a^{qt+r} \equiv a^q t a^r \equiv (a^t)^q a^r \equiv a^r \pmod{n}$. By definition of the order of a modulo n , we can easily check that $r = 0$. Therefore $t|m$.

$\Rightarrow$ Assume that $m = tq$. Then $a^m \equiv a^{tq} \equiv (a^t)^q \equiv 1^q \equiv 1 \pmod{n}$. Now we are done. □

Corollary 1.52. Let a and n be positive integers. Let t be the order of a mod n . Then $1, a, \dots, a^{t-1}$ are distinct integers modulo n .

Proof. Assume that $0 \leq s < r < t$. If $a^r \equiv a^s \pmod{n}$, then $t|r - s$. Since $0 < r - s < t$, it is a contradiction. $\square$

Exercise 1.53. Calculate $\phi(2010)$.

Exercise 1.54. Give an example s.t. $\phi(mn) \neq \phi(m)\phi(n)$.

Exercise 1.55. Find the least positive integer a such that for every $n \in \mathbb{N}$, $\phi(n) \neq a$.

Exercise 1.56. Find the number of invertible a ($1 \leq a \leq 524$) modulo 524.

Exercise 1.57. Find the order of 7 modulo 18.

1.9 Primitive root of mod n

Definition 1.58. The integer a is called a primitive root of mod n if the order of a mod n is $\phi(n)$.

Example 1.59. 1. The order of 2 mod 5 is 4. Note that $\phi(5) = 5 - 1 = 4$. Therefore 2 is a primitive root of mod 5.

2. The order of 2 mod 7 is 3. Note that $\phi(7) = 7 - 1 = 6$. Therefore 2 is not a primitive root of mod 7. Since 3 is prime, we can observe that 3 is the order of 2 modulo 7.

3. There is no primitive root of mod 8. Because $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$.

Theorem 1.60. There is a primitive root of modulo n if and only if $n = 2, 4, p^k$ or $2p^k$ for some prime p and positive integer k .

(Artin conjecture) Let $a(a > 1)$ be an integer. There are infinitely many primes p s.t. a is a primitive root of modulo p .

Theorem 1.61 (Euler's theorem). Let a and n be positive integers. Suppose that $\gcd(a, n) = 1$. Then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Let's confirm the theorem for mod 7.

	x	x^2	x^3	x^4	x^5	x^6
1	1	1	1	1	1	1
2	2	4	1	2	4	1
3	3	2	6	4	5	1
4	4	2	1	4	2	1
5	5	4	6	2	3	1
6	6	1	6	1	6	1

Then 3 and 5 are primitive root of modulo 7.

Theorem 1.62 (Fermat little theorem). *If p is a prime and a is an integer with $\gcd(a, p) = 1$, then $a^{p-1} \equiv 1 \pmod{p}$.*

Proof. By the Euler's theorem, we can easily check that $a^{\phi(p)} \equiv a^{p-1} \equiv 1 \pmod{p}$. $\square$

Example 1.63. 1. Find $3^{205} \pmod{101}$. By the Fermat little theorem, $3^{\phi(101)} \equiv 3^{100} \equiv 1 \pmod{101}$. Therefore

$$3^{205} \equiv 3^{200} 3^5 \equiv 3^5 \equiv 243 \equiv 38 \pmod{101}.$$

2. Find $5^{10000} \pmod{18}$. Note that $\phi(18) = 6$. By Euler's theorem, $5^6 \equiv 1 \pmod{18}$. Therefore

$$5^{10000} \equiv 5^4 \equiv 25^2 \equiv 7^2 \equiv 13 \pmod{18}.$$

Exercise 1.64. *If p is a prime and a is an integer, then $a^p \equiv a \pmod{p}$.*

Exercise 1.65. *Find all primitive roots of mod 12.*

Exercise 1.66. *Find all primitive roots of mod 10.*

Exercise 1.67. *Find $6^{10004} \pmod{11}$.*

Exercise 1.68. *Find $2^{10000} \pmod{13}$.*

Exercise 1.69. *Find the number of a ($1 \leq a \leq 18$) such that a is invertible of mod 18.*

2 Cryptology

2.1 Diffie-Hellman key exchange

Diffie-Hellman key exchange is the algorithm such that two people communicate through the public channel and share their secret keys. Let's call above two people Alice and Bob for convenience, simply A and B. Public channel means "Anyone know messages sent by Alice and Bob". How can Alice and Bob establish a secret key? We know that For $a, N \in \mathbb{Z}_{>0}$, it is easy to calculate $a^N \pmod{n}$ and for a given $y \pmod{n}$ and $a \in \mathbb{Z}_{>0}$, it is very hard to find N such that $a^N \equiv y \pmod{n}$.

Definition 2.1. For a given $y \pmod{n}$ and $a \in \mathbb{Z}_{>0}$, the task of finding N such that $a^N \equiv y \pmod{n}$ is called the discrete logarithm problem.

Example 2.2. Let $2^N \equiv 3 \pmod{11}$. Find N .

N	1	2	3	4	5	6	7	8	...
2^N	2	4	8	5	10	9	7	3	...

Therefore $N = 8$.

To get N , compute all $a^m(\text{mod } n)$ for $0 \leq m \leq n$. Suppose that size of n , N about 10^{100} , then we have to check 10^{100} possible N .

(Algorithm of Diffie-Hellman key exchange)

Goal : A and B share secret key k such that not easily calculated by an eavesdropper.

1. A and B publicly choose a large prime p and a base a .
2. A secretly choose a number s and sends $a^s(\text{mod } p)$ to B.
3. B secretly choose a number t and sends $a^t(\text{mod } p)$ to A.
4. A secretly calculate $k \equiv (a^t)^s(\text{mod } p)$ and B secretly calculate $k \equiv (a^s)^t(\text{mod } p)$.

Eavesdropper(Eve) only knows a , p , $a^s(\text{mod } p)$, and $a^t(\text{mod } p)$. Therefore, to know k , Eve need to know s or t . Then Eve need to solve the discrete logarithm problem. But it is not easy to proof.

Example 2.3. Assume that $a = 2$ and $p = 11$. A take $s = 4$ and B take $t = 8$. Let's calculate the Diffie-Hellman key exchange algorithm.

1. A sends $2^4 \equiv 5(\text{mod } 11)$ to B.
2. B sends $2^8 \equiv 3(\text{mod } 11)$ to A.
3. A receives 3 from B, calculate $3^s \equiv 3^4 \equiv 4(\text{mod } 11)$.
4. B receives 5 from A, calculate $5^t \equiv 5^8 \equiv 4(\text{mod } 11)$.

Then secret key is 4.

Exercise 2.4. Find N such that $3^N \equiv 9(\text{mod } 13)$.

Exercise 2.5. Give the Diffie-Hellman key exchange algorithm.

Exercise 2.6. Let $p = 13$ and base 3. A take 6 and B take 5. In this case, by following the Diffie-Hellman key exchange algorithm step by step. Calculate the secret key.

2.2 RSA

In 1977, Rivest, Shadmir, and Adleman introduced the RSA algorithm. It is an encryption scheme which is a way of sending message so that no unauthorized person can read it. Before RSA, encryption and decryption with the same key. But, in RSA, one key is used for encryption and different

key is used for decryption. Note that RSA is a public-key cryptosystem, anyone with a public key can encrypt a message, but a private key is required for decryption and difficult to know the private key through the public key.

(Algorithm of RSA)

1. Take large primes p, q (each with at least 100 decimal digits)
 2. Compute $N = pq$, $\phi(N) = (p-1)(q-1)$. And randomly choose an integer e such that $\gcd(e, \phi(N)) = 1$.
 3. Using Euclidean algorithm, Compute $d \equiv e^{-1}(\text{mod } \phi(N))$.
 4. Announce public key (N, e) .
- **Public key** : (N, e)
 - **message** : $x(0 < x < N)$
 - **Encryption** : $x^e(\text{mod } N)$
 - **Decryption** : $y^d(\text{mod } N)$ for y received message i.e., $y \equiv x^e(\text{mod } N)$

Note that range of the message is not a big deal. Because computer files consist of numbers and a large number can split into blocks. We can easily check that

$$y^d \equiv (x^e)^d \equiv x^{ed} \equiv x^{1+t\phi(N)} \equiv x \cdot (x^{\phi(N)})^t \equiv x(\text{mod } N).$$

Example 2.7. Take $p = 5, q = 11$. Let $N = pq = 55$. Then $\phi(N) = 40$. Let public key $(N, e) = (55, 3)$. Then $d \equiv 27(\text{mod } 40)$. Choose $x = 4$. Then encryption of 4 is $x^e \equiv 4^3 \equiv 64 \equiv 9(\text{mod } 55)$. And decryption of 9 is $9^{27} \equiv 4(\text{mod } 55)$.

Exercise 2.8. Consider the RSA algorithm. Let $(N, e) = (323, 11)$. Suppose that the message is 316.

1. Find the secret key.
2. Find the encrypted message.
3. Explain how to decrypt the encrypted message in ii).

Exercise 2.9. Let $N = pq$. Estimate $\frac{\phi(N)}{N}$.

3 Continued Fractions

We'll be using [1, Chapter 15] as a base to build on in Chapter 15 of this book.

3.1 Finite continued fractions

For example, $\frac{61}{14}$ can be expressed as $4 + \frac{1}{2 + \frac{1}{1 + \frac{1}{4}}}$.

$$\begin{aligned}\frac{61}{14} &= 4 + \frac{5}{14} \\ &= 4 + \frac{1}{\frac{14}{5}} \\ &= 4 + \frac{1}{2 + \frac{4}{5}} \\ &= 4 + \frac{1}{2 + \frac{1}{\frac{5}{4}}} \\ &= 4 + \frac{1}{2 + \frac{1}{1 + \frac{1}{4}}}\end{aligned}$$

Definition 3.1. 1. By a **finite continued fraction** is meant a fraction of the form

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \cdots \frac{1}{a_{n-1} + \frac{1}{a_n}}}}$$

where $a_0 \in \mathbb{R}$ and $a_i \in \mathbb{R}_{>0}$ for all $1 \leq i \leq n$.

2. Such a fraction is called simple if all of a_i are integers.

Theorem 3.1. *Any rational number can be written as a finite simple continued fraction.*

The key point of proof of above theorem is the Euclid's algorithm.

Proof. Let r be an arbitrary rational number $\frac{a}{b}$ ($b > 0$). By Euclid's algorithm, we have the equations

$$\begin{aligned} a &= ba_0 + r_1, 0 < r_1 < b \\ b &= r_1a_1 + r_2, 0 < r_2 < r_1 \\ r_1 &= r_2a_2 + r_3, 0 < r_3 < r_2 \\ &\dots \\ r_{n-2} &= r_{n-1}a_{n-1} + r_n, 0 < r_n < r_{n-1} \\ r_{n-1} &= r_na_n + 0. \end{aligned}$$

Then we know that

$$\begin{aligned} \frac{a}{b} &= a_0 + \frac{r_1}{b} = a_0 + \frac{1}{\frac{b}{r_1}} \\ \frac{b}{r_1} &= a_1 + \frac{r_2}{r_1} = a_1 + \frac{1}{\frac{r_1}{r_2}} \\ &\dots \\ \frac{r_{n-1}}{r_n} &= a_n. \end{aligned}$$

In above result, we can get the formula

$$\frac{a}{b} = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots + \frac{1}{a_{n-1} + \frac{1}{a_n}}}}.$$

Now we are done. □

Example 3.2. Let us represent $\frac{19}{51}$ as a simple continued fraction. By Euclid's algorithm, we have the equations

$$51 = 2 \cdot 19 + 13, 19 = 1 \cdot 13 + 6, 13 = 2 \cdot 6 + 1, 6 = 6 \cdot 1 + 0.$$

Making the simple continued fraction using above formula, it is seen that

$$\begin{aligned} \frac{19}{51} &= \frac{1}{\frac{51}{19}} \\ &= \frac{1}{2 + \frac{13}{19}} \\ &= \frac{1}{2 + \frac{1}{\frac{19}{13}}} \\ &= \frac{1}{2 + \frac{1}{1 + \frac{6}{13}}} \\ &= \frac{1}{2 + \frac{1}{1 + \frac{1}{\frac{13}{6}}}} \\ &= \frac{1}{2 + \frac{1}{1 + \frac{1}{2 + \frac{1}{6}}}} \end{aligned}$$

Definition 3.3. The symbol $[a_0; a_1, \dots, a_n]$ denotes a finite continued fraction

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \dots + \frac{1}{a_{n-1} + \frac{1}{a_n}}}}.$$

For example, in the case of $\frac{19}{51}$,

$$\frac{19}{51} = [0; 2, 1, 2, 6] = [0; 2, 1, 2, 5, 1].$$

Remark that a simple continued fraction $[a_0; a_1, \dots, a_n]$ is same as $[a_0; a_1, \dots, a_n - 1, 1]$.

[Simple continued fraction algorithm]

- **Input :** $\frac{a}{b}$ a rational number ($b > 0$)
- **Output :** $[a_0; a_1, \dots, a_n]$ a simple continued fraction equal to $\frac{a}{b}$
 - (a) Let $a = a_0b + r_1$, $b = a_1r_1 + r_2$.
 - (b) If $r_i \neq 0$, then $r_{i-1} = a_i r_i + r_{i+1}$.
 - (c) If $r_{i-1} = 0$, then terminate this algorithm.

Exercise 3.4. Express each number as a simple continued fraction $\frac{172}{51}, \frac{71}{55}$.

Exercise 3.5. By Euclid's algorithm, find $[a_0; a_1, \dots, a_n] = \frac{118}{303}$.

Exercise 3.6. Find the rational numbers represented by the following simple continued fractions,

$$[-2; 2, 4, 6, 8], [0; 1, 2, 3, 4, 3, 2, 1]$$

Definition 3.7. For $1 \leq k \leq n$, the k th convergent of $[a_0; a_1, \dots, a_n]$ is $[a_0; a_1, \dots, a_k]$, denoted by C_k . For convenience, $C_0 = a_0$.

For example, $\frac{19}{51} = [0; 2, 1, 2, 6]$, the successive convergents are

$$C_0 = 0$$

$$C_1 = [0; 2] = 0 + \frac{1}{2} = \frac{1}{2}$$

$$C_2 = [0; 2, 1] = 0 + \frac{1}{2 + \frac{1}{1}} = \frac{1}{3}$$

$$C_3 = [0; 2, 1, 2] = 0 + \frac{1}{2 + \frac{1}{1 + \frac{1}{2}}} = \frac{3}{8}$$

$$C_4 = [0; 2, 1, 2, 6] = \frac{19}{51}.$$

Remark that $C_k = [a_0; a_1, \dots, a_k]$ and $C_{k+1} = [a_0; a_1, \dots, a_k + \frac{1}{a_{k+1}}] = [a_0; a_1, \dots, a_{k+1}]$.
How to compute the value of C_k ? Define the number p_k, q_k for $[a_0; a_1, \dots, a_n]$ ($0 \leq k < n$),

$$p_0 = a_0, q_0 = 1$$

$$p_1 = a_1 a_0 + 1, q_1 = a_1$$

$$\dots$$

$$p_k = a_k p_{k-1} + p_{k-2}, q_k = a_k q_{k-1} + q_{k-2}.$$

A direct computation shows that the first few convergents of $[a_0; a_1, \dots, a_n]$ are

$$C_0 = a_0 = \frac{a_0}{1} = \frac{p_0}{q_0}$$

$$C_1 = a_0 + \frac{1}{a_1} = \frac{a_1 a_0 + 1}{a_1} = \frac{p_1}{q_1}$$

$$C_2 = a_0 + \frac{1}{a_1 + \frac{1}{a_2}} = \frac{a_2(a_1 a_0 + 1) + a_0}{a_2 a_1 + 1} = \frac{p_2}{q_2}.$$

For convenience, define $p_{-2} = 0, q_{-2} = 1, p_{-1} = 1, q_{-1} = 0$. Then $p_k = a_k p_{k-1} + p_{k-2}, q_k = a_k q_{k-1} + q_{k-2}$ for all $k \geq 0$.

Theorem 3.8. *The k th convergent of the simple continued fraction $[a_0; a_1, \dots, a_n]$ has the value*

$$C_k = \frac{p_k}{q_k}, \quad 0 \leq k \leq n.$$

Proof. It is true for $k = 0, 1, 2$. Assume that $k = m$ for $2 \leq m < n$ theorem is true for $k = m$. Then

$$C_m = \frac{p_m}{q_m} = \frac{a_m p_{m-1} + p_{m-2}}{a_m q_{m-1} + q_{m-2}}. \text{ Then}$$

$$\begin{aligned} C_{m+1} &= [a_0; a_1, \dots, a_{m+1}] \\ &= \left[a_0; a_1, \dots, a_m + \frac{1}{a_{m+1}} \right] \\ &= \frac{\left(a_m + \frac{1}{a_{m+1}} \right) p_{m-1} + p_{m-2}}{\left(a_m + \frac{1}{a_{m+1}} \right) q_{m-1} + q_{m-2}} \\ &= \frac{a_{m+1}(a_m p_{m-1} + p_{m-2}) + p_{m-1}}{a_{m+1}(a_m q_{m-1} + q_{m-2}) + q_{m-1}} \\ &= \frac{a_{m+1} p_m + p_{m-1}}{a_{m+1} q_m + q_{m-1}} \\ &= \frac{p_{m+1}}{q_{m+1}}. \end{aligned}$$

□

Let us see how this works in a specific example, $\frac{19}{51} = [0; 2, 1, 2, 6]$

$$\begin{aligned} p_0 &= 0, q_0 = 1 \\ p_1 &= 0 \cdot 2 + 1 = 1, q_1 = 2 \\ p_2 &= 1 \cdot 1 + 0 = 1, q_2 = 1 \cdot 2 + 1 = 3 \\ p_3 &= 2 \cdot 1 + 1 = 3, q_3 = 2 \cdot 3 + 2 = 8 \\ p_4 &= 6 \cdot 3 + 1 = 19, q_4 = 6 \cdot 8 + 3 = 51 \end{aligned}$$

It says that the convergents of $[0; 2, 1, 2, 6]$ are

$$C_0 = \frac{p_0}{q_0} = 0, C_1 = \frac{p_1}{q_1} = \frac{1}{2}, C_2 = \frac{p_2}{q_2} = \frac{1}{3}, C_3 = \frac{p_3}{q_3} = \frac{3}{8}, C_4 = \frac{p_4}{q_4} = \frac{19}{51}.$$

Theorem 3.9. If $C_k = \frac{p_k}{q_k}$ is the k th convergent of the finite simple continued fraction $[a_0; a_1, \dots, a_n]$, then

$$p_k q_{k-1} - q_k p_{k-1} = (-1)^{k-1}, 1 \leq k \leq n.$$

Proof.

□

Corollary 3.10. For $1 \leq k \leq n$, p_k and q_k are relatively prime.

Proof. Assume that $d|q_k, d|p_k$. We know that $p_k q_{k-1} - q_k p_{k-1} = (-1)^{k-1}$. Then $d|L.H.S$. So $d|(-1)^{k-1}$. Therefore $d = 1$. Now we are done. □

Summary, k th convergent of $[a_0; a_1, \dots, a_n]$ is $[a_0; a_1, \dots, a_k] = \frac{p_k}{q_k}$. And $\gcd(p_k, q_k) = 1$.

For example, consider the continued fraction $[0; 1, 1, \dots, 1]$. Note that $p_{-2} = 0, q_{-2} = 1, p_{-1} = 1, q_{-1} = 0$. Then $p_k = a_k p_{k-1} + p_{k-2}, q_k = a_k q_{k-1} + q_{k-2}$ for all k . Then $C_0 = \frac{0}{1}, C_1 = \frac{1}{1}, C_2 = \frac{1}{2}, C_3 = \frac{2}{3}, \dots$. The numerator of the k th convergent C_k is $p_k = 1 \cdot p_{k-1} + p_{k-2} = p_{k-1} + p_{k-2}$ and the denominator is $q_k = 1 \cdot q_{k-1} + q_{k-2} = q_{k-1} + q_{k-2}$. Define u_k be the k th Fibonacci number.

Then $C_k = \frac{u_k}{u_{k-1}}$. By corollary 3.10, we know that $u_{k+1}u_{k-1} - u_k^2 = (-1)^{k-1}$ for integer k .

For example of theorem 15. 8, let us solve the linear Diophantine equation $172x + 20y = 1000$. We need to solve the $43x + 5y = 250$ ($\because \gcd(43, 5) = 1$). By Euclid's algorithm, we have the formula

$$43 = 8 \cdot 5 + 3, 5 = 1 \cdot 3 + 2, 3 = 1 \cdot 2 + 1, 2 = 2 \cdot 1.$$

Therefore

$$\begin{aligned} \frac{43}{5} &= [8; 1, 1, 2] \\ &= 8 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2}}} \end{aligned}$$

We know that $q_2 = 2$ and $p_2 = 17$ and $p_k q_{k-1} - q_k p_{k-1} = (-1)^{k-1}$. Therefore $43 \cdot 2 - 5 \cdot 17 = (-1)^2 = 1$. So we have an integral solution $(x, y) = (2 \cdot 250, 17 \cdot 250) = (500, 4250)$ of $172x + 20y = 1000$.

Lemma 3.11. If q_k is the denominator of the k th convergent C_k of the simple continued fraction $[a_0; a_1, \dots, a_n]$, then $q_{k-1} \leq q_k$ for $1 \leq k \leq n$, when $k > 1$.

Proof. Assume that $k = 1$. Then $q_0 = 1$, $q_1 = a_1 \geq 1$. Therefore $q_0 \leq q_1$. Assume that it is true for $k = m$ ($1 \leq m < n$). Then

$$q_{m+1} = a_m q_m + a_{m-1} > a_{m+1} q_m \geq 1 \cdot q_m = q_m.$$

□

Theorem 3.12. 1. $C_0 < C_2 < C_4 < \dots$

2. $C_1 > C_3 > C_5 > \dots$

3. Every convergent with an odd script is greater than every convergent with an even script.

Proof.

□

Exercise 3.13. (*) Let $C_k = \frac{p_k}{q_k}$ be the k th convergent of $[1; 2, 3, \dots, n, n+1]$. Prove that $p_n = np_{n-1} + np_{n-2} + (n-1)p_{n-3} + \dots + 3p_1 + 2p_0 + p_0 + 1$.

Exercise 3.14. Let $C_k = \frac{p_k}{q_k}$ be the k th convergent of the simple continued fraction $[a_0; a_1, \dots, a_n]$. Prove that $q_k \geq 2^{\frac{k-1}{2}}$, $2 \leq k \leq n$.

Exercise 3.15. Compute the convergents of the $[1; 2, 3, 3, 2, 1]$ and $[-3; 1, 1]$.

Exercise 3.16. Find the simple continued fraction representation of 3.1416, and that of 3.14159.

Exercise 3.17. If $C_k = \frac{p_k}{q_k}$ is the k th convergent of the simple continued fraction $[a_0; a_1, \dots, a_n]$ and $a_0 > 0$, show that

$$\frac{p_k}{p_{k-1}} = [a_k; a_{k-1}, \dots, a_1, a_0], \quad \frac{q_k}{q_{k-1}} = [a_k; a_{k-1}, \dots, a_1]$$

Exercise 3.18. By means of continued fractions determine the general solutions of each of the following Diophantine equations:

1. $19x + 51y = 1$.

2. $18x + 5y = 24$.

Exercise 3.19. Verify Theorem 3.12 for the simple continued fraction $[1; 1, 1, 1, 1, 1, 1, 1]$.

3.2 Infinite continued fractions

Theorem 3.20 (Uniqueness of a simple continued fraction). *If $[a_0; a_1, \dots, a_n] = [b_0; b_1, \dots, b_m]$ for $a_i, b_i > 1$ for all $i \geq 1$ and $a_n, b_m > 1$, then $n = m$ and $a_i = b_i$ for all $i \leq n$.*

Corollary 3.21. *There are exactly two ways of writing any rational number as a finite simple continued fraction.*

For example, $[7] = [6; 1]$, $[1; 2, 2, 2] = [1; 2, 2, 1, 1]$, and $[1; 1; 1; 1; 2] = [1; 1, 1, 1, 1]$.

Assume that $x \in \mathbb{R}$ irrational number. Then $\exists a_i \in \mathbb{Z}_{>0} (i \geq 1)$ such that

$$x = \lim_{n \rightarrow \infty} [a_0; a_1, \dots, a_n].$$

Let $[a_0; a_1, \dots, a_n, \dots]$ denote $\lim_{n \rightarrow \infty} [a_0; a_1, \dots, a_n]$.

[Computing a simple continued fraction]

- **Input** : a real number x
 - **Output** : $[a_0; a_1, \dots, a_n, \dots]$ such that $x = \lim_{k \rightarrow \infty} C_k$
1. $x_0 = x$.
 2. Let a_m be the integral part of x_m , it means $[x_m]$.
 3. $\xi_m := x_m - a_m$.
 4. If ξ_m , then set $\frac{1}{\xi_m}$ as x_{m+1} and go back to step 2.
 5. If $\xi_m = 0$, then terminate this algorithm.

For example, $\pi = [3; 7, 15, 1, 292, \dots]$.

Definition 3.22. The continued fraction $[a_0; a_1, \dots, a_n, b_1, \dots, b_m, b_1, \dots, b_m, \dots]$ denoted by $[a_0; a_1, \dots, a_n, \overline{b_1, \dots, b_m}]$.

Example 3.23. Let $x = [3; 6, \overline{1, 4}]$. Define $y = [\overline{1; 4}]$. Then $x = [3; 6, y]$ and $y = [1; 4, \overline{1, 4}] = [1; 4, y]$. Therefore we know that

$$y = 1 + \frac{1}{4 + \frac{1}{y}}.$$

Since $y > 0$, $y = \frac{1 + \sqrt{2}}{2}$. Therefore

$$x = 3 + \frac{1}{6 + \frac{1}{\frac{1 + \sqrt{2}}{2}}} = \frac{14 - \sqrt{2}}{4}.$$

Exercise 3.24. Let $[a_0; a_1, \dots, a_k]$ be a simple continued fraction. Given recursive relations for p_k and q_k ($0 \leq k \leq n$) such that

$$[a_0; a_1, \dots, a_k] = \frac{p_k}{q_k}.$$

Exercise 3.25. Evaluate the following

1. $[2; 3]$.
2. $[2; 1, 2, 1]$.

Exercise 3.26. Prove that $\forall n \in \mathbb{Z}_{>0}$

1. $\sqrt{n^2 + 1} = [n; 2n]$.
2. $\sqrt{n^2 + 2} = [n; n, 2n]$.
3. $\sqrt{n^2 + 2n} = [n; 1, 2n]$.

Then we have a question, what is the best approximation to a given number? We need to define the best approximation.

Definition 3.27. A rational number $\frac{p}{q}$ is the best approximation to a real number x if $\left|x - \frac{p}{q}\right| \leq |x - r|$ for every rational number r such that $|\text{denominator } r| \leq |q|$.

Theorem 3.28. Assume that $x = [a_0; a_1, \dots]$ a simple continued fraction. Then $q_n \geq n$ when $n \geq 3$ if $x = [a_0; a_1, \dots, a_m]$, then $n \leq m$.

Proof. We know that $q_0 = 1$, $q_1 = a_1$. For $n \geq 2$, we know that

$$q_n = a_n q_{n-1} + q_{n-2} \geq a_n q_{n-1} + 1 \leq q_{n-1} + 1.$$

Therefore $q_2 \geq q_1 + 1 = 2$. Assume that for $k = m$, $q_m \geq m$. Then $q_{m+1} \geq q_m + 1 \geq m + 1$. So the inequality is also true for $k = m + 1$. Therefore $q_n \geq n$ for all integer n . $\square$

Theorem 3.29. A finite simple continued fraction $[a_0; a_1, \dots, a_n]$ is equal to

$$a_0 + \frac{1}{q_1 q_0} - \frac{1}{q_2 q_1} + \dots + (-1)^n \frac{1}{q_n q_{n-1}}.$$

Proof. Note that $p_m q_{m-1} - q_m p_{m-1} = (-1)^{m-1}$ for all integer m . Then

$$\begin{aligned} C_n &= \frac{p_n}{q_n} \\ &= \frac{p_n q_{n-1}}{q_n q_{n-1}} \\ &= \frac{q_n p_{n-1} + (-1)^{n-1}}{q_n q_{n-1}} \\ &= \frac{p_{n-1}}{q_{n-1}} + \frac{(-1)^{n-1}}{q_n q_{n-1}} \\ &= \frac{p_{n-2}}{q_{n-1}} + \frac{(-1)^{n-2}}{q_{n-1} q_{n-2}} + \frac{(-1)^{n-1}}{q_n q_{n-1}} \\ &= \dots \\ &= \frac{p_0}{q_0} + \sum_{i=1}^n (-1)^{i-1} \frac{1}{q_i q_{i-1}} \\ &= a_0 + \sum_{i=1}^n (-1)^{i-1} \frac{1}{q_i q_{i-1}} \end{aligned}$$

□

Theorem 3.30. For any real number x with convergents $\frac{p_m}{q_m}$ for integer m ,

$$\left| x - \frac{p_m}{q_m} \right| < \frac{1}{q_{m+1} q_m}.$$

Proof. 추후 추가

□

Theorem 3.31. Let x be a real number and simple continued fraction $x = [a_0; a_1, \dots, a_n]$ (n can be ∞). For $m < n$ and $1 \leq b \leq q_m$ ($b \in \mathbb{Z}$), every rational number $\frac{a}{b}$ ($a \in \mathbb{Z}$) satisfies

$$\left| x - \frac{p_m}{q_m} \right| \leq \left| x - \frac{a}{b} \right|,$$

in other words $\frac{p_m}{q_m}$ is the best approximation of x .

Example 3.32. To calculate the simple continued fraction of $\sqrt{2}$, first note that $1 < \sqrt{2} < 2$. So $a_0 = 1$. Then

$$\sqrt{2} = 1 + (\sqrt{2} - 1) = 1 + \frac{1}{\sqrt{2} + 1} = 1 + \frac{1}{2 + (\sqrt{2} - 1)}.$$

Therefore $a_n = 2$ for all integer n . Therefore $\sqrt{2} = [1; \overline{2}]$. The following table is value of p_n, q_n for all $n \geq 0$. Therefore $\frac{41}{29}$ is an best approximation of $\sqrt{2}$.

n	0	1	2	3	4	$\dots$
p_n	1	$1 \cdot 2 + 1 = 3$	$3 \cdot 2 + 1 = 7$	17	41	$\dots$
q_n	1	2	5	12	29	$\dots$

Exercise 3.33. Let $x = [1; 3, 1, 5, 1, 7, 1, \dots]$. Find the best rational approximation $\frac{a}{b}$ with

1. denominator $b \leq 23$.
2. (*) denominator $b \leq 239$.

Exercise 3.34. 1. By using the continued fraction, find a rational number $\frac{q}{p}$ such that

$$\left| \sqrt{15} - \frac{q}{p} \right| < \frac{1}{50^2}.$$

(Hint : Let $\frac{p_n}{q_n}$ be the n th convergent of the simple continued fraction of a real number x . Then

$$\left| x - \frac{p_m}{q_m} \right| < \frac{1}{q_{m+1}q_m})$$

2. (*) Among k th convergents of $\sqrt{15}$, find a rational number that approximates $\sqrt{15}$ accuracy to Four decimal places.

Exercise 3.35. Prove that

$$\left| \frac{1 + \sqrt{10}}{3} - \frac{18}{13} \right| < \frac{1}{2 \cdot 13^2} (*)$$

and prove that $\frac{18}{13}$ is a convergent to $\frac{1 + \sqrt{10}}{3}$.

4 Basic of Quantum computer

Quantum computers have the advantages of analog computers and the advantages of digital computers. The advantage of digital computers is that they use digital data, which consists of 0 and 1.

It makes it convenient to input and output data. The analog computers use analog data. It has the advantage of being efficiency of operations.

For example, consider $1.21212137 + 0.12121211$. Let A be the 1.21212137 and B be the 0.12121211 . First, describe an operation on an analog computer. Note that Kirchhoff's current law states that the current flowing into a node must be equal to the current flowing out of it. So If two wires with current A and one with current B meet at a point as a single wire, then the current after they meet at the point is $A + B$. Second, describe an operation on a digital computer. A 9-digit number expressed in bits is 30 digits. Even bit additive operations require a cost. In this case, an analog computer is more efficient than a digital computer. However, the downside of analog computers is that data is difficult to input and output, and computation requires an ideal environment.

The advantage of digital computers is that they are like particles, and the advantage of analog computers is that they are like waves. Quantum computers use superposition and collapse to exploit the advantages of waves and particles. More explicitly, a superposition state is collapsed through a process called measurement.

4.1 one-qubit

A bit is the unit information in classical computer consisting of 0 and 1.

Definition 4.1. A qubit is the unit information in quantum computer consisting of unit vector of

$$\mathbb{C}^2 = \mathbb{C} \times \mathbb{C} = \{(x, y) | x, y \in \mathbb{C}\}.$$

Let's denote $(1, 0), (0, 1)$ by e_1, e_2 , respectively. It means that $(x, y) = xe_1 + ye_2$.

If we measure a qubit (x, y) , then we measure 1 with probability $|x|^2$ and 0 with probability $|y|^2$, i.e., this means the superposition of 0 and 1 in classical computer.

Moreover, denote $(1, 0), (0, 1)$ by $|0\rangle, |1\rangle$, respectively. For example, $(2, 3) = 2|0\rangle + 3|1\rangle$. Every one qubit can be expressed as the following form $a|0\rangle + b|1\rangle$ such that

$$|a|^2 + |b|^2 = 1, (a, b \in \mathbb{C}).$$

	classical computer	quantum computer
unit information	a bit 0, 1	a qubit, a unit vector in $\mathbb{C}^2$ ($ae_1 + be_2, a ^2 + b ^2 = 1$)

Exercise 4.2. Fill the blanks

“the basis idea of quantum computing is based on combining advantage of $\boxed{(a)}$ and $\boxed{(b)}$.”

Exercise 4.3. Explain

1. *advantages of analog computer*
2. *advantages of digital computer*
3. *disadvantage of analog computer*

Exercise 4.4. *What is a qubit?*

Exercise 4.5. *Express the following vectors as a linear combination of $|0\rangle$ and $|1\rangle$,*

1. $(2, 4)$
2. $(7, 2)$
3. $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$

4.2 n -qubit

Definition 4.6. 1. Two-qubit is defined by one qubit $\otimes$ one qubit. It means that two-qubit is the form of

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$$

where $a, b, c, d \in \mathbb{C}$, $|a|^2 + |b|^2 = 1$ and $|c|^2 + |d|^2 = 1$. Like polynomial expansion, the above tensor can also be computed as follows:

$$(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|0\rangle \otimes |0\rangle + ad|0\rangle \otimes |1\rangle + bc|1\rangle \otimes |0\rangle + bd|1\rangle \otimes |1\rangle.$$

Simply,

2. n -qubit is the form of

$$(a_1|0\rangle + b_1|1\rangle) \otimes \cdots \otimes (a_n|0\rangle + b_n|1\rangle)$$

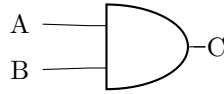
where $a_i, b_i \in \mathbb{C}$, $|a_i|^2 + |b_i|^2 = 1$ for all $i \leq n$.

Note that $|ac|^2 + |ad|^2 + |bc|^2 + |bd|^2 = (|a|^2 + |b|^2)(|c|^2 + |d|^2) = 1$. We simply denote $|ab\rangle := |a\rangle \otimes |b\rangle$.

4.3 Classical gate

A classical gate is a fundamental building block in digital logic circuits used to perform basic logical operations on binary data. Assume that $T : \text{true}$ corresponds 1 and $F : \text{False}$ corresponds 0.

Definition 4.7. At the "And gate" and "Or gate", the number of input data is 2, which we will denote as A and B . And the number of output data is 1, which we will denote as C . It looks like the figure below.



1. The **And gate** is a basic digital logical gate behaves according to the truth table.

A	B	$C(A \wedge B)$
T	T	T
T	F	F
F	T	F
F	F	F

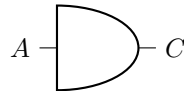
And gate can be viewed as multiplication modulo 2, it means that $AB \equiv C(mod2)$.

2. The **Or gate** is a basic digital logical gate behaves according to the truth table.

A	B	$C(A \vee B)$
T	T	T
T	F	T
F	T	T
F	F	F

Or gate can be viewed as $(A + 1)(B + 1) + 1 \equiv C(mod2)$.

3. At the "Not gate", the number of input data is 1, which we will denote as A . And the number of output data is 1, which we will denote as C . It looks like the figure below.



The **Not gate** is a basic digital logical gate behaves according to the truth table.

A	C
T	F
F	T

In fact, any gates can be implemented by using "and", "or" and "not" gate. It is universal gate theorem.

4.4 Quantum gate

Definition 4.8. 1. $A \in M_{n \times n}(\mathbb{C})$ is unitary if $A\bar{A}^T = I_n$, i.e., $A^{-1} = \bar{A}^T$

2. A quantum gate for 1-qubit is a unitary 2×2 matrix.

For example,

$$U(x_1|0\rangle + x_2|1\rangle) = U \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = x'_1|0\rangle + x'_2|1\rangle$$

Example 4.9. Let $|0\rangle = (1, 0)$ and $|1\rangle = (0, 1)$. Then $U := \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is quantum “not” gate.

$$U|0\rangle = U \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

$$U|1\rangle = U \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

Note that n -qubit is the form of $(a_1|0\rangle + b_1|1\rangle) \otimes \cdots \otimes (a_n|0\rangle + b_n|1\rangle) \in \mathbb{C}^{2^n}$. Since quantum gates are unitary matrix, we know that quantum gates send n -qubit to n -qubit (same dimension).

Example 4.10. 1. Controlled not gate is a quantum logical gate behaves according to the truth table.

A	B	A'	B'
$ 0\rangle$	$ 0\rangle$	$ 0\rangle$	$ 0\rangle$
$ 0\rangle$	$ 1\rangle$	$ 0\rangle$	$ 1\rangle$
$ 1\rangle$	$ 0\rangle$	$ 1\rangle$	$ 1\rangle$
$ 1\rangle$	$ 1\rangle$	$ 1\rangle$	$ 0\rangle$

Let $|0\rangle|0\rangle = (1, 0, 0, 0)$, $|0\rangle|1\rangle = (0, 1, 0, 0)$, $|1\rangle|0\rangle = (0, 0, 1, 0)$ and $|1\rangle|1\rangle = (0, 0, 0, 1)$. Then the matrix corresponding to the CNOT gate is as follows:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

2. X -gate and Z -gate are example of single gate such that

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{X} \longrightarrow \beta|0\rangle + \alpha|1\rangle$$

$$\alpha|0\rangle + \beta|1\rangle \longrightarrow \boxed{Z} \longrightarrow \alpha|0\rangle - \beta|1\rangle$$

Compute to find the matrix corresponding to the X gate and Z gate.

In fact, every quantum gate is implemented by using 1-qubit gates and CNOT gate.

Exercise 4.11. Give the definition of a unitary matrix.

Exercise 4.12. Fill the blank

1. A quantum gate can be express as a $\boxed{(a)}$ matrix.
2. Every quantum gate can be implemented by $\boxed{(b)}$ and $\boxed{(c)}$

Exercise 4.13. Let $(x|0\rangle + y|1\rangle) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)$ be a 2-qubit where $(|x|^2 + |y|^2 = 1)$ and U be CNOT gate. Find $U(x|0\rangle + y|1\rangle) \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right)$.

Exercise 4.14. Let U be a quantum gate for 1-qubit such that

$$\begin{aligned} |0\rangle &\xrightarrow{U} \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \\ |1\rangle &\xrightarrow{U} \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle. \end{aligned}$$

Find the matrix corresponding to U .

Exercise 4.15. Let U be the quantum gate for 2-qubit such that

$$\begin{array}{cc} (0,0) & (0,1) \\ (0,1) & \xrightarrow{U} (1,0) \\ (1,0) & (1,1) \\ (1,1) & (0,0) \end{array}$$

Let $|0\rangle|0\rangle := (1, 0, 0, 0)$, $|0\rangle|1\rangle := (0, 1, 0, 0)$, $|1\rangle|0\rangle := (0, 0, 1, 0)$ and $|1\rangle|1\rangle := (0, 0, 0, 1)$. Find a matrix corresponding to U .

Exercise 4.16. Prove that matrices corresponding the X gate and Z gate.

4.5 measurement

In this subsection, let $H := \mathbb{C}^n$ be a vector space with inner product.

Definition 4.17. A **state** is a unit vector v in H .

Let $\Omega \in M_n(\mathbb{C})$ be a Hermitian matrix, i.e., $\Omega = (\overline{\Omega})^t$. Note that a Hermitian matrix is diagonalizable and every eigenvalue of Hermitian matrix is real. Let $\{v_i\}_{1 \leq i \leq n}$ be the set of orthonormal basis of Ω such that

$$\Omega v_i = \lambda_i v_i \text{ for all } i.$$

For convenience, assume that all λ_i 's are distinct. Note that $v = \sum_{i=1}^n a_i v_i$ such that $\sum_{i=1}^n |a_i|^2 = 1$. When the state v is measured using the operator Ω , the following results are obtained: the eigenvalue λ_i can be measured with a probability of $|a_i|^2$ and the state v collapses to the state v_i .

$$v \xrightarrow[\text{measure}]{\Omega} \begin{array}{l} \text{measurement } \lambda_i \text{ with prob } |a_i|^2 \\ \text{new state } v_i \end{array}$$

Let's better understand measurement through an example.

Example 4.18. Let $v = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \in \mathbb{C}^2$ and $\Omega = \begin{pmatrix} -2 & \\ & 1 \end{pmatrix}$. Then eigenvectors of Ω are $(1, 0)$ and $(0, 1)$. Then eigenvalues of $(1, 0)$ and $(0, 1)$ are -2 and 1 , respectively. Therefore

$$v \xrightarrow[\text{measure}]{\Omega} \begin{array}{l} \text{measurement } -2 \text{ with prob } (1/2)^2 \text{ and new state } (1, 0) \\ \text{measurement } 1 \text{ with prob } (\sqrt{3}/2)^2 \text{ and new state } (0, 1) \end{array}$$

Example 4.19. Let v be a state in one qubit. Let Ω be a hermitian matrix such that

$$\begin{aligned} \Omega \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) &= \lambda_1 \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) \\ \Omega \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) &= \lambda_2 \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right). \end{aligned}$$

Let $v_1 := \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ and $v_2 := \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$. Assume that $v = \frac{1}{\sqrt{3}}|0\rangle + \frac{\sqrt{2}}{\sqrt{3}}|1\rangle$. Then $v = \frac{1+\sqrt{2}}{\sqrt{6}}v_1 + \frac{1-\sqrt{2}}{\sqrt{6}}v_2$. Therefore

$$v \xrightarrow[\text{measure}]{\Omega} \begin{array}{l} \text{measurement } \lambda_1 \text{ with prob } ((1+\sqrt{2})/\sqrt{6})^2 \text{ and new state } v_1 \\ \text{measurement } \lambda_2 \text{ with prob } ((1-\sqrt{2})/\sqrt{6})^2 \text{ and new state } v_2 \end{array}$$

Example 4.20. Let Ω be a hermitian matrix such that

$$\Omega|0\rangle = \lambda_1|0\rangle, \Omega|1\rangle = \lambda_2|1\rangle.$$

Let $v = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ be a state. Then

$$v \xrightarrow[\text{measure}]{\Omega} \begin{array}{l} \text{measurement } \lambda_1 \text{ with prob } 1/2 \text{ and new state } |0\rangle \\ \text{measurement } \lambda_2 \text{ with prob } 1/2 \text{ and new state } |1\rangle \end{array}$$

Exercise 4.21. Give the definition of Hermitian matrix.

Exercise 4.22. Let v be a state. Explain “measure” for v .

1. Determine the properties of the matrix Ω related to measurement.
2. Find the measurement when v is measured using Ω .
3. Determine the state of v corresponding to each measurement value when v is measured using Ω .

Exercise 4.23. Let $v = \frac{1}{\sqrt{5}}|0\rangle + \frac{2}{\sqrt{5}}|1\rangle$ and $\Omega = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

1. What is probability with measurement 2?
2. What is probability with measurement 1?

4.6 Hadamard and CNOT gate

The Hadamard gate is a 1-qubit gate such that

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{H} \alpha\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + \beta\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Let $|+\rangle$ denote $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle$ denote $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

Then simply,

$$|0\rangle \xrightarrow{H} |+\rangle, |1\rangle \xrightarrow{H} |-\rangle.$$

Define $|0\rangle := (1, 0)$ and $|1\rangle := (0, 1)$. Then the matrix corresponding to the Hadamard gate is as follows

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Let us represent various gates using symbols. The following symbols represent the controlled NOT gate and the SWAP gate. Note that $B \oplus A \equiv B + A \pmod{2}$.

Figure file's [Figure.1] corresponds to the symbol for a CNOT gate, and [Figure.2] corresponds to the symbol for a SWAP gate.

Exercise 4.24. *Explain the Hadamard gate.*

Exercise 4.25. *Based on [Figure.3] and [Figure.4] in the Figure file, determine the respective $|A\rangle|B\rangle$ states corresponding to each Figure.*

References

- [1] D. Burton, *EBOOK: Elementary Number Theory*. McGraw Hill, 2010.